# Problem 10: Step size

### Setup

All the runs below use the step response from problem 9.4 (PD gains 100 and 30, simulated for 4 s). The default plant step is 0.1 ms and the default controller period is 1 ms; I changed one at a time.

### How the two step sizes relate

The plant step is how often the physics is integrated, and the controller period defines how often a new torque is computed (with torque held constant between updates). Therefore, the controller period should be a whole number of plant steps, so that the plant moves smoothly under a constant torque, and the controller see a fresh state every time it is run.

### Plant time step and error

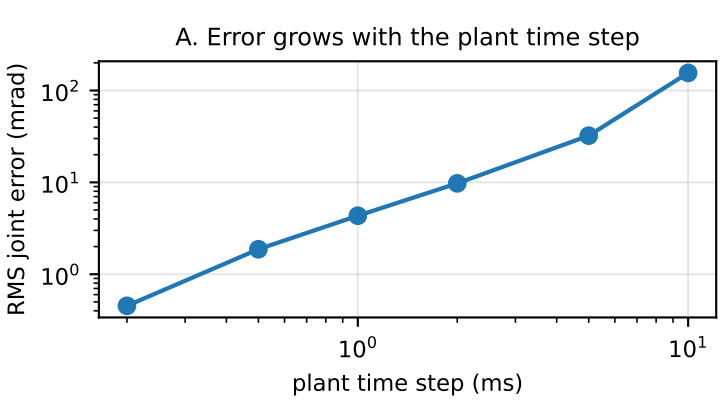
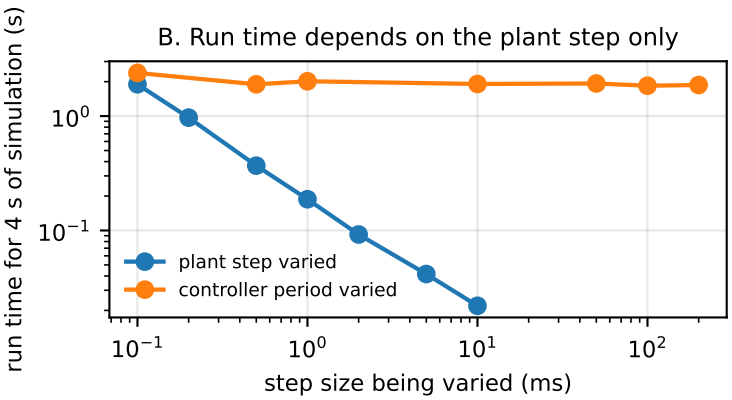
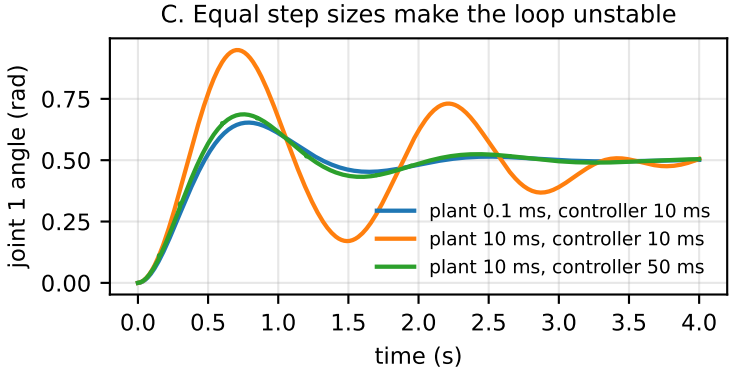
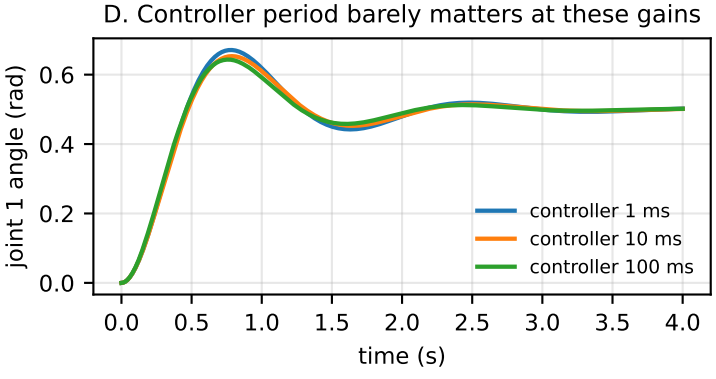
I varied plant time step and computed the RMS joint error of each run (panel A). The error is roughly proportional to the step size (doubling the step doubles the error), which makes sense given that we are using a first-order integrator.

### Wall-clock time

Since the plant step directly determines how many physics steps there are, this clearly means that runtime decreases as the plant time step increases (panel B). The controller period, on the other hand, has no measurable effect: with the plant at 0.1 ms, every controller period from 0.1 ms to 200 ms took about 2 s (the flat line in panel B). Therefore, the plant step is the only knob that makes the simulation faster, and a 1 ms step gives a 10x speedup for a few milliradians of error.

### Failure mode

If the plant step is set equal to the controller period (both 10 ms), the arm never settles and just keeps oscillating (panel C). Part of this is presumably that a 10 ms integration step is coarse (panel A shows the error growing steadily with the step size), but that can't be the whole story: the same 10 ms plant step with a 50 ms controller period settles fine, and setting both steps to 0.1 ms, the finest step in the sweep, gives the same instability. So the two step sizes being equal is a problem in itself. The reason is that Drake's discrete plant reports its state with a delay of one plant step. Normally this doesn't matter since the controller period is much longer than one plant step, but when the two are equal the position the controller reads is a full period old. This means the velocity estimate (which the PD controller gets by differencing consecutive positions) is also a full period out of date, and so the damping term ends up pushing in the wrong direction and adds energy instead of removing it.

### Controller period

For the gains in 9.4 the controller period hardly matters, since 1 ms, 10 ms and 100 ms all give nearly the same response (panel D). This makes sense because the closed loop takes about a second to settle, so even a 100 ms update is fast compared to the motion it is controlling. Therefore, as long as the controller period is several plant steps long (to avoid the failure mode above), there is little to gain from running the controller any faster than the response time of the loop requires.